

UniTS Anomaly Ratio Sweep

STPSat4-HRR | Generated 2026-03-12 10:53

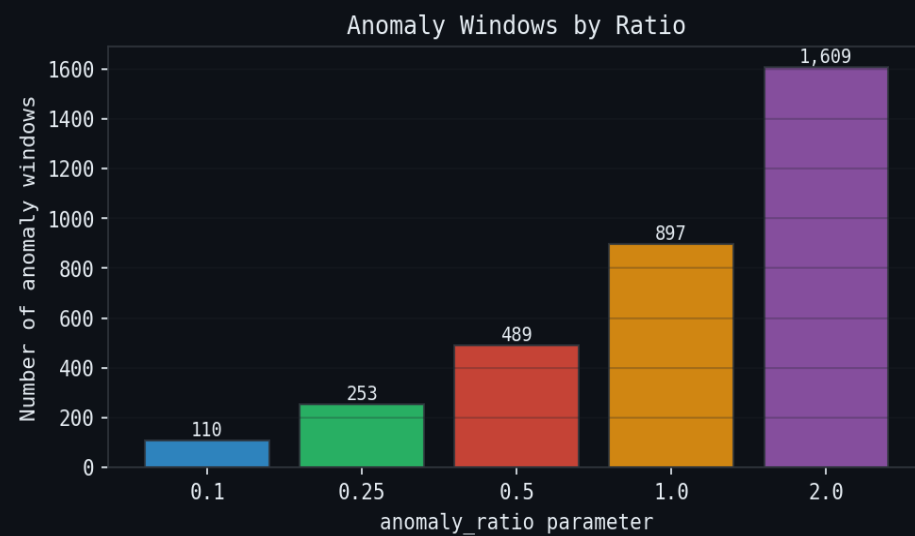
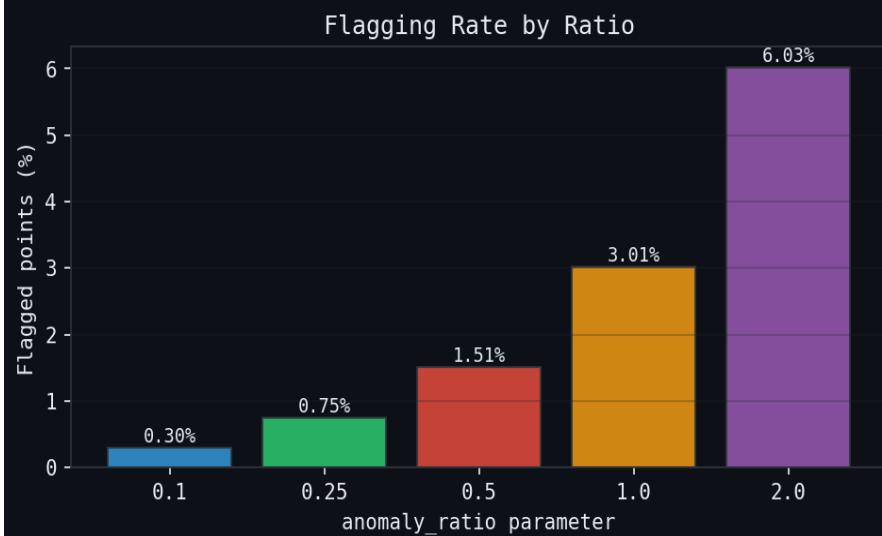
Results by Ratio

anomaly_ratio	Threshold	Flagged Points	Flag Rate	Windows
0.1	0.470593	131	0.300%	110
0.25	0.289360	329	0.753%	253
0.5	0.212198	658	1.506%	489
1.0	0.156527	1,316	3.013%	897
2.0	0.108175	2,632	6.026%	1,609

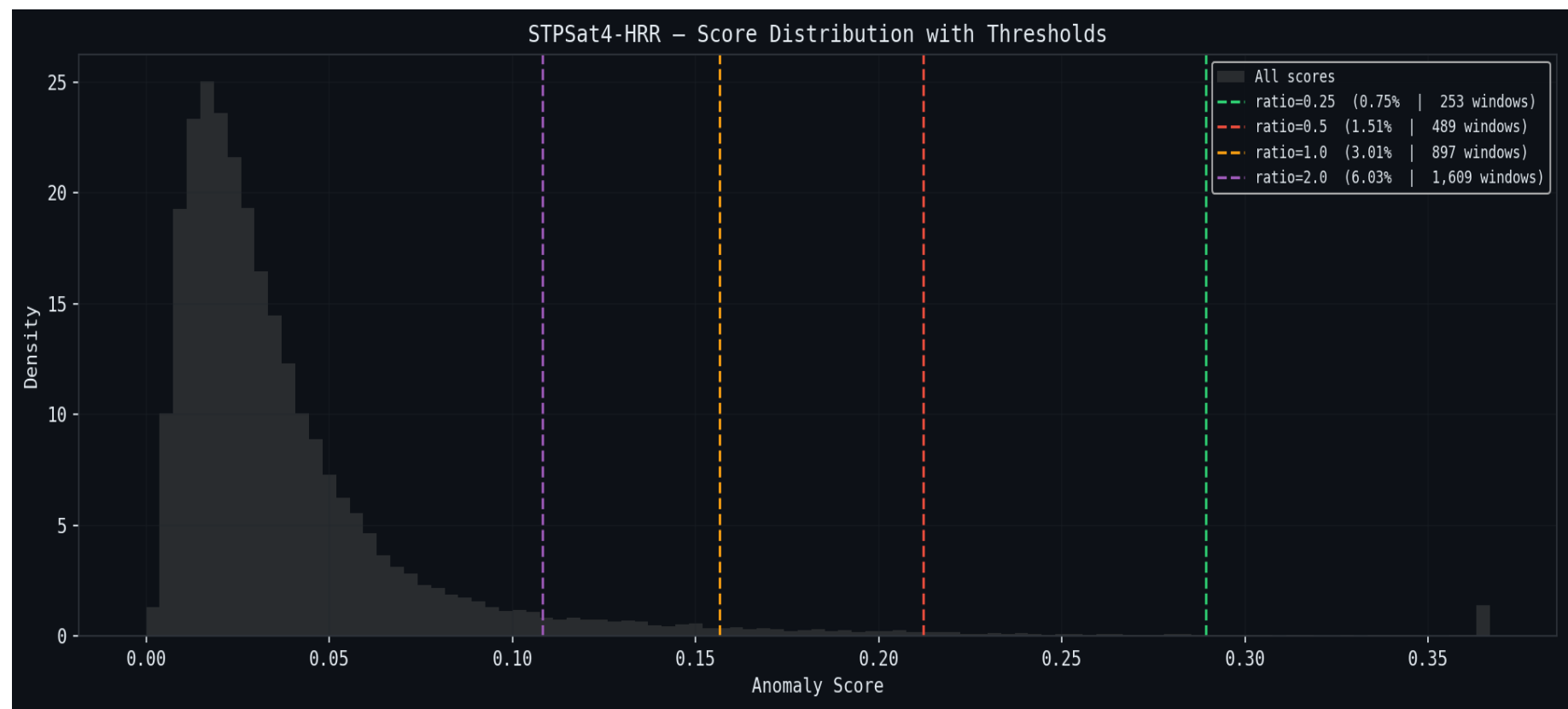
anomaly_ratio controls the detection threshold: UniTS flags the top ratio% of reconstruction error scores as anomalies. Lower values = stricter threshold = fewer flags. The threshold column shows the exact score cutoff for each ratio.

Flagging Rate & Window Count

STPSat4-HRR – Ratio Sweep Summary



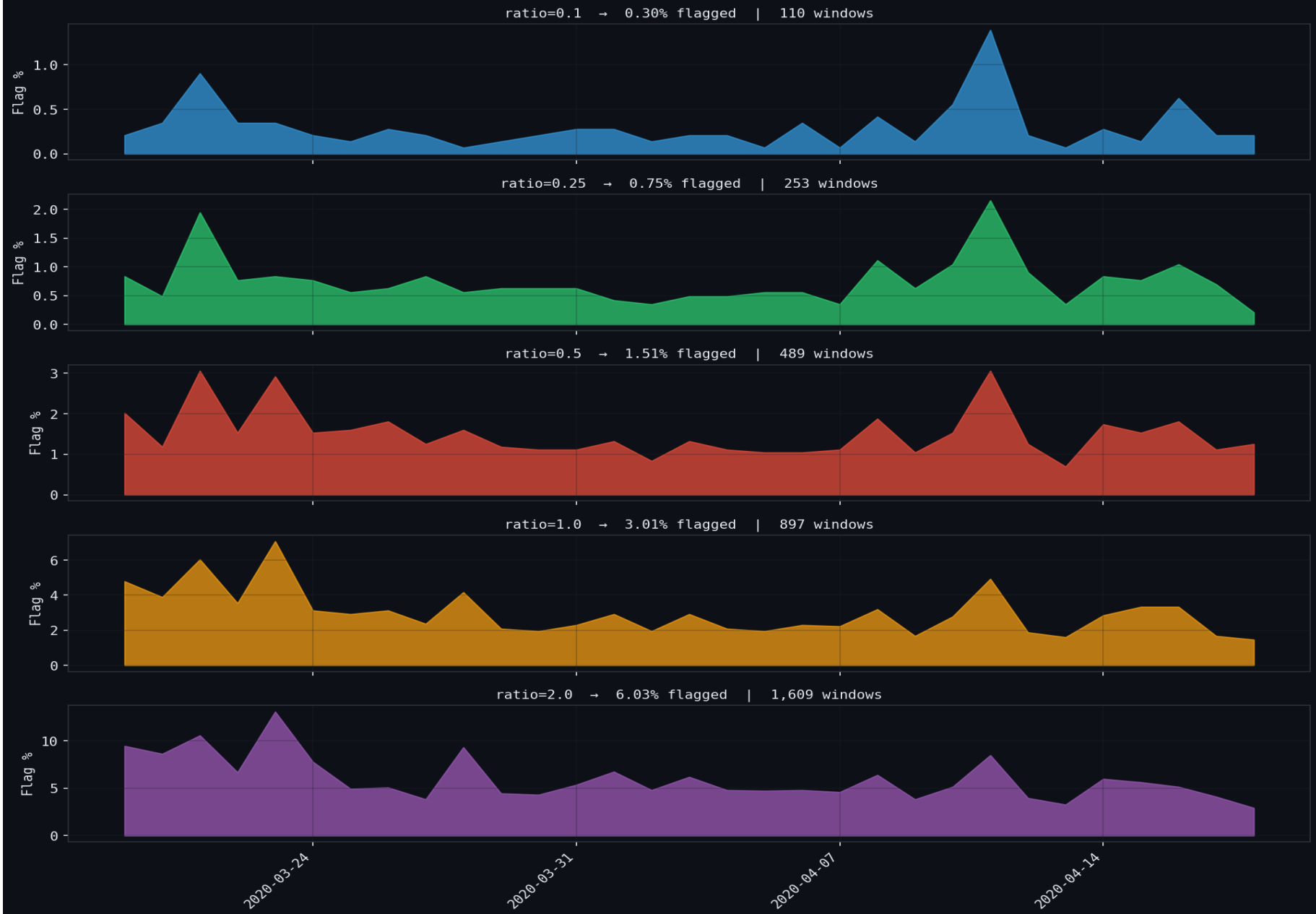
Score Distribution with Thresholds



Dashed vertical lines show where each ratio sets its threshold. Choose a ratio whose threshold sits clearly in the tail of the distribution rather than in the main body of normal scores.

Daily Flagging Rate by Ratio

STPSat4-HRR — Daily Flagging Rate by Ratio



A realistic anomaly rate for a nominal satellite is typically 0.1–1%. Look for a ratio that shows isolated spikes rather than sustained elevated flagging.